



Security Overview of Ecopath with Ecosim (EwE)

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Overview

Ecopath with Ecosim (EwE) is a sandboxed, desktop-only ecosystem modeling application used worldwide for marine resource assessments, food web modeling, and ecosystem-based fisheries management.

EwE runs locally on a user's computer. It does not require internet access, does not collect or transmit data, and requires no elevated system permissions. For over 30 years, the EwE desktop software has been and is being used by academic, government, and intergovernmental institutions globally, including FAO, NOAA, CSIRO, CSIC, CEFAS, EU-JRC, and various universities and ministries.

Security posture

Category	EwE Status
Execution	Desktop application; runs in user space; no kernel access
Networking	No internet access, no telemetry, no sockets, no background services
Data Handling	Does not process personal or sensitive data
Encryption	No built-in cryptographic modules; not applicable
Installation	Windows installer signed by Ecopath International Initiative (EII)
Dependencies	No bundled third-party runtime environments; limited to .NET/Core if used
Code Signing	All official EwE builds are digitally signed
AV/Threat Scan	Verified clean on VirusTotal and major antivirus engines

Applicability of standards and regulations

Standard / Regulation	Relevance to EwE
SOC 2 / SOC 3	Not applicable – EwE is not a service; no operational infrastructure
ISO/IEC 27001	Not applicable – EwE is not a managed IT system or data processor
FIPS 140-2	Not applicable – EwE performs no cryptographic functions
GDPR	Not applicable – EwE does not process or store personal data



Independent Assurance

We are currently exploring third-party code or binary audits to formally verify EwE's security-neutral posture. In the meantime, EwE is:

- Regularly scanned via VirusTotal.
- Packaged via signed and hashed installers.
- Used under long-standing institutional software governance in over 150 countries by an estimated 10.000 users.

Summary

EwE is a safe, self-contained, non-networked ecological modelling application. It poses negligible risk to institutional IT environments and requires no special exception handling in modern desktop security policies.

If your institution requires further assurances, we will be happy to:

- Provide signed VirusTotal scan results.
- Share internal security architecture documentation.
- Coordinate with your IT team for sandbox inspection or testing.